首都师范大学
CAPITAL NORMAL UNIVERSITY

电路分析

电阻电路的等效化简

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一 → 等效及等效化简
（Equivalency & equivalent simplification）

等效

二端网络N_1

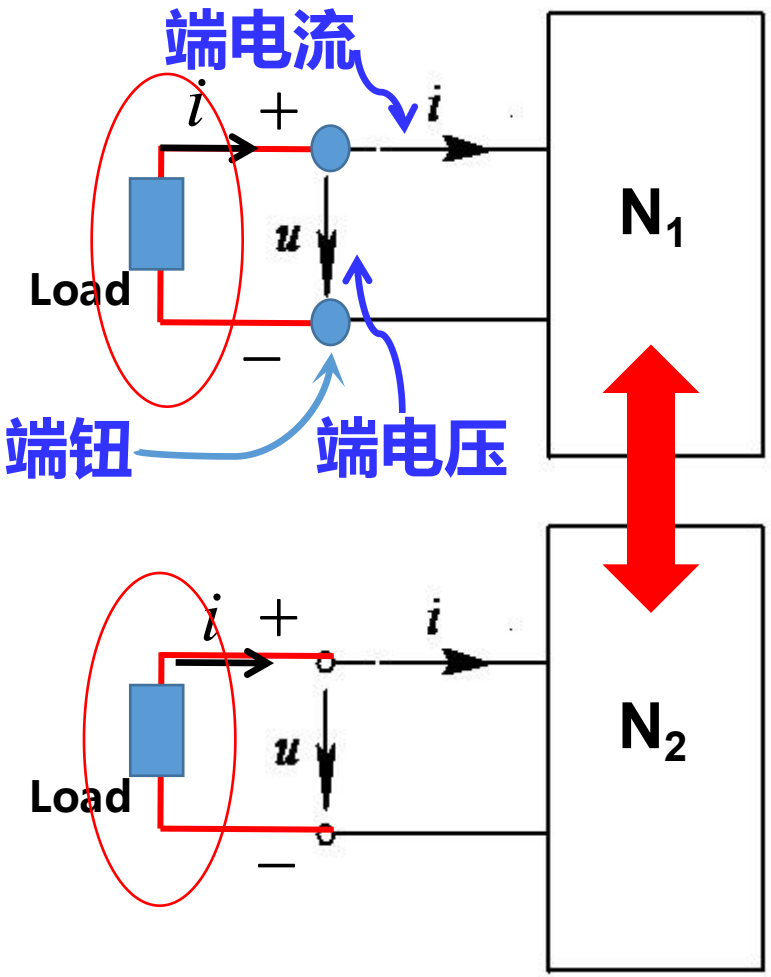
端钮伏安关系 $u = f_1(i)$

二端网络N_2 $u = f_2(i)$

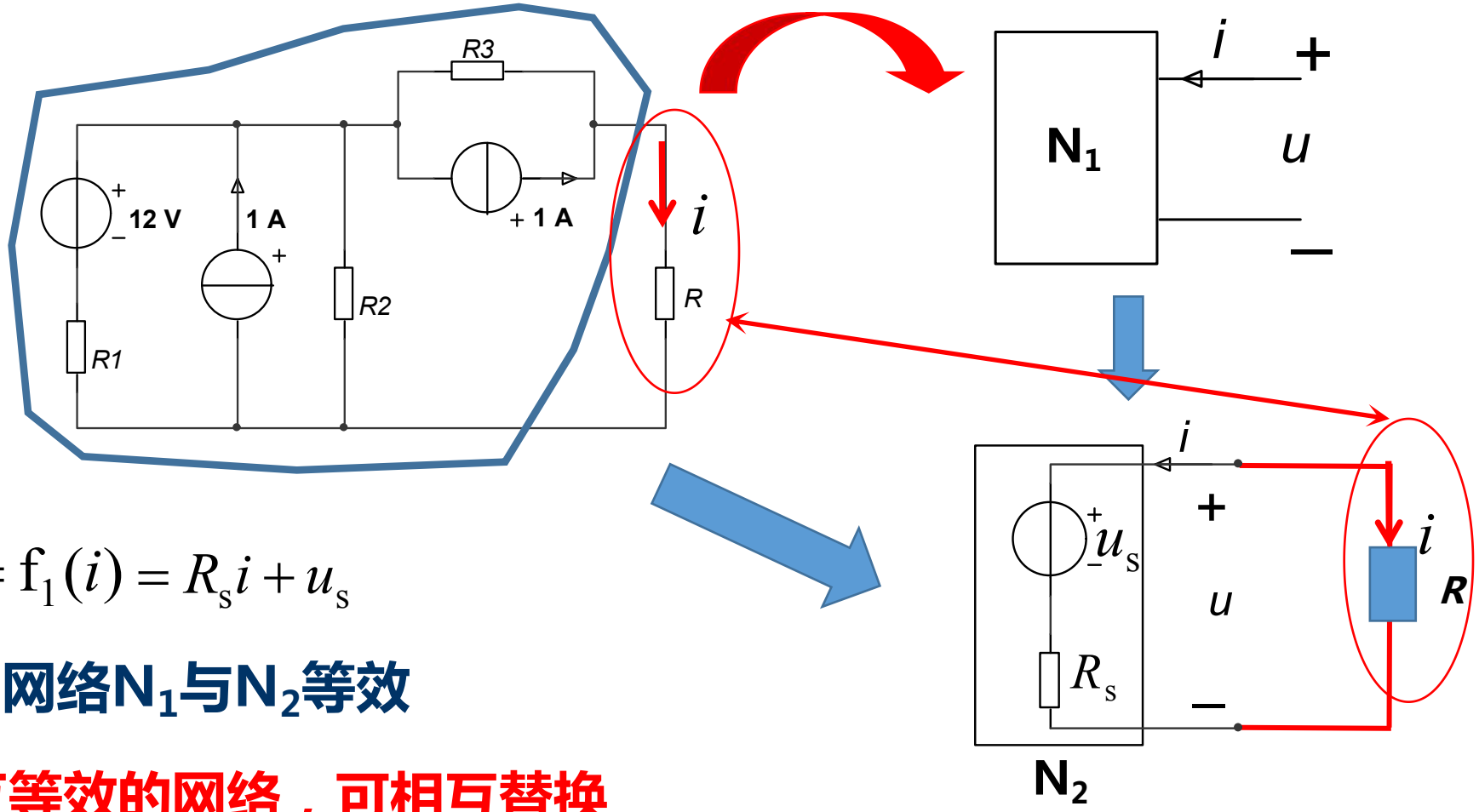
如果 $u = f_1(i) = f_2(i)$

端对外电路而言，N_1 和N_2 是等效的

等效是对针外电路。内电路不等效。



等效化简



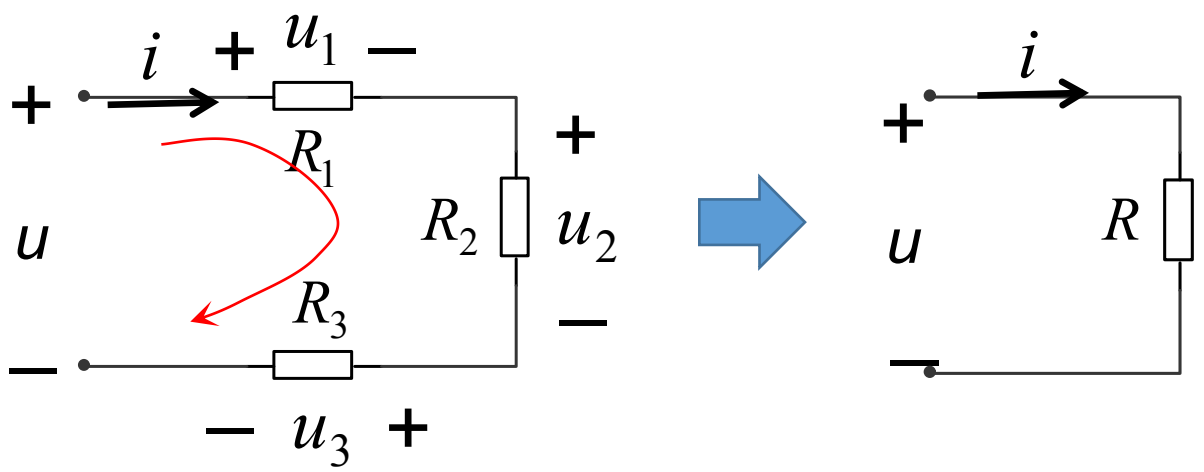
$$u = f_1(i) = R_s i + u_s$$

二端网络N_1与N_2等效

相互等效的网络，可相互替换

二 二端电阻网络的等效（equivalent simplification for two-terminal resistor network）

1 串联电阻网络的等效（resistors in serial）



$$u = f(i) = u_1 + u_2 + u_3 = R_1 i + R_2 i + R_3 i$$

$$= (R_1 + R_2 + R_3) i = Ri$$

$$R = R_1 + R_2 + R_3$$

n个电阻串联

$$R = R_1 + R_2 + R_3 + \cdots + R_n = \sum_{j=1}^{n} R_j$$

串联电阻的分压

$$u_1 = R_1 i = R_1 \cdot \frac{u}{R} = \frac{R_1}{R} u$$

$$u_2 = \frac{R_2}{R} u$$

$$u_3 = \frac{R_3}{R} u$$

$$u_1 : u_2 : u_3 = R_1 : R_2 : R_3$$

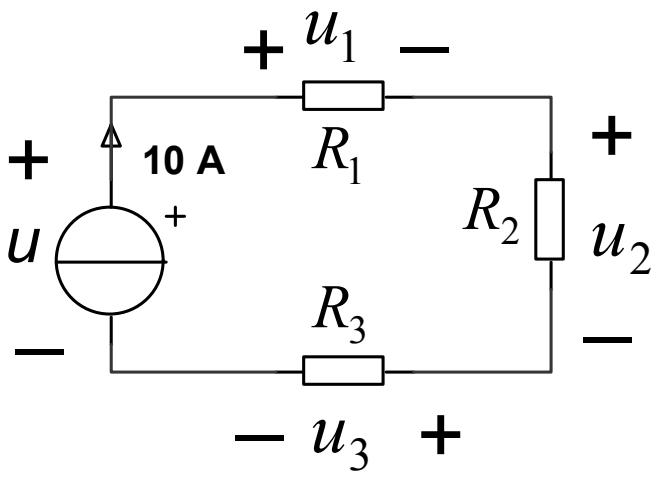
n个串联电阻的分压

$$u_j = \frac{R_j}{R} \cdot u$$

二 二端电阻网络的等效
(equivalent simplification for two-terminal resistor network)

2 串联电阻网络的等效例题

例1，R_1=2Ω， R_2=5Ω ,R_3=1Ω ，求各电阻两端的电压。



解： $R = R_1 + R_2 + R_3$

$$= 2 + 5 + 1 = 8\ (\Omega)$$

$$u = 10R = 10 \times 8 = 80\ (\mathrm{V})$$

$$u_1 = \frac{R_1}{R}u = \frac{2}{8} \times 80 = 20\ (\mathrm{V})$$

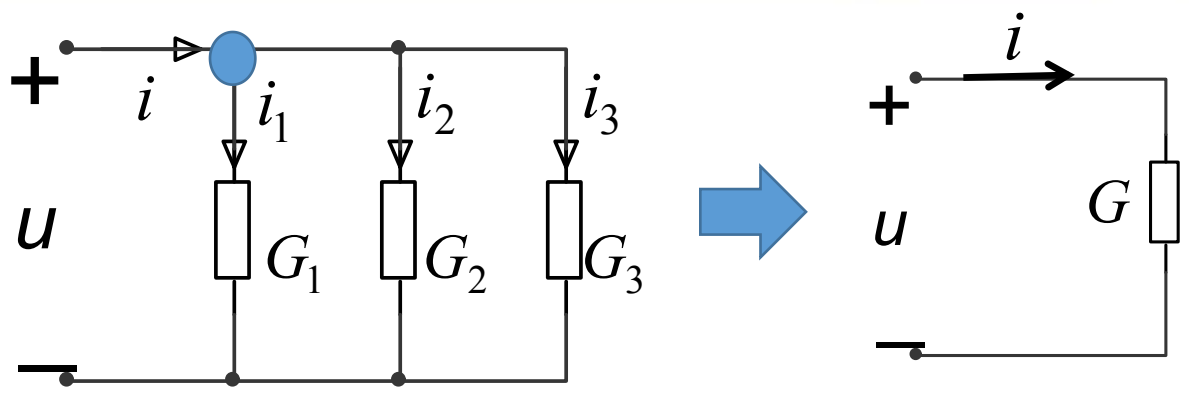
$$u_2 = \frac{R_2}{R}u = \frac{5}{8} \times 80 = 50\ (\mathrm{V})$$

$$u_3 = \frac{R_3}{R}u = \frac{1}{8} \times 80 = 10\ (\mathrm{V})$$

$$u_1 : u_2 : u_3 = R_1 : R_2 : R_3 = 2:5:1$$

二 二端电阻网络的等效（equivalent simplification for two-terminal resistor network）

3 并联电阻网络的等效（resistors in parallel）



$$i = f(u) = i_1 + i_2 + i_3 = G_1 u + G_2 u + G_3 u$$

$$= (G_1 + G_2 + G_3) u = G u$$

$$G = G_1 + G_2 + G_3$$

n个电阻并联

$$G = G_1 + G_2 + G_3 + \cdots + G_n$$

$$= \sum_{j=1}^{n} G_j$$

$$R = R_1 // R_2 = \frac{R_1 R_2}{R_1 + R_2}$$

并联电阻的分流

$$i_1 = G_1 u = G_1 \cdot \frac{i}{G} = \frac{G_1}{G} i$$

$$i_2 = \frac{G_2}{G} i$$

$$i_3 = \frac{G_3}{G} i$$

n个电阻并联分流

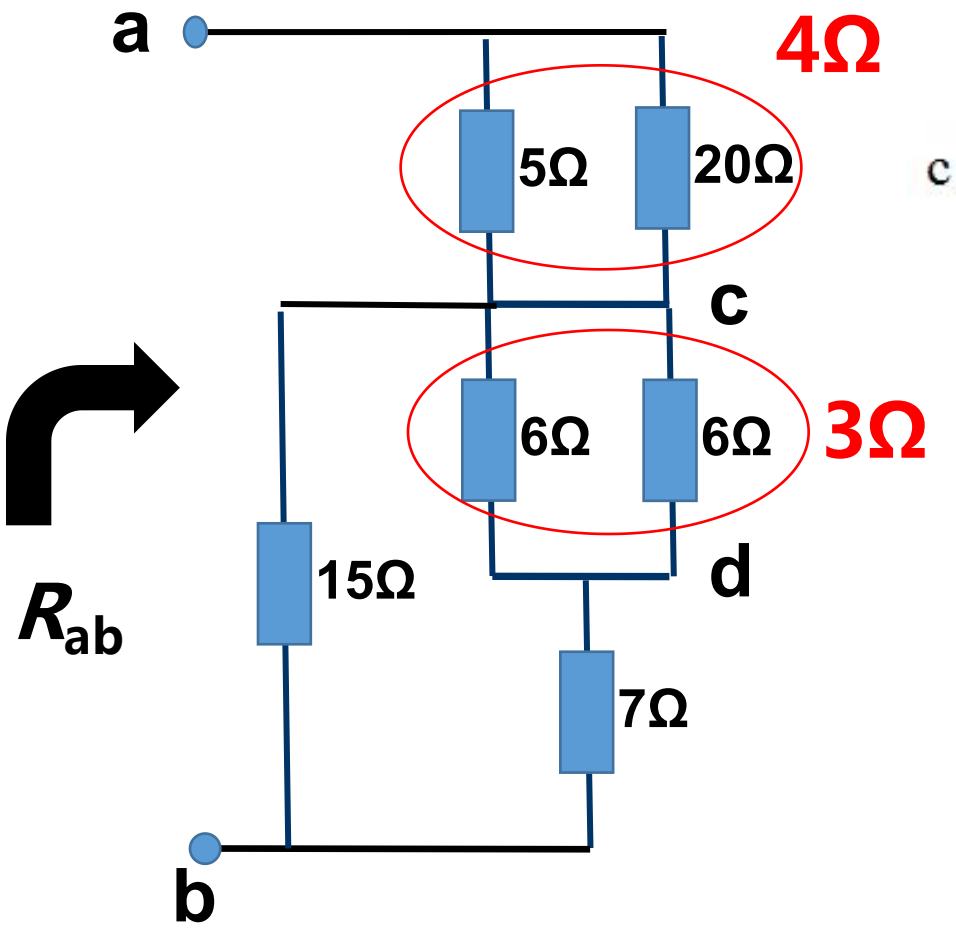
$$i_j = \frac{G_j}{G} i$$

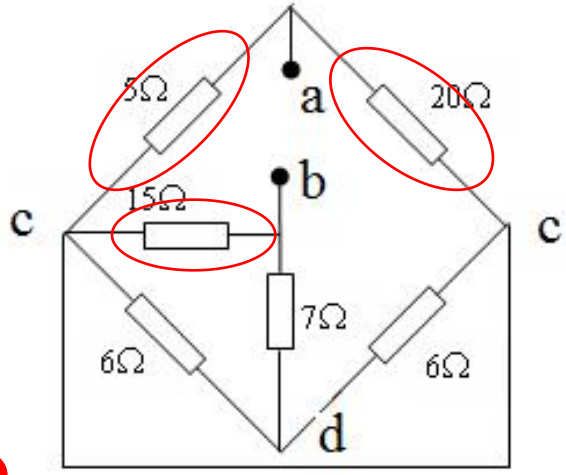
$$i_1 = \frac{R_2}{R_1 + R_2} \cdot i \qquad i_2 = \frac{R_1}{R_1 + R_2} \cdot i$$

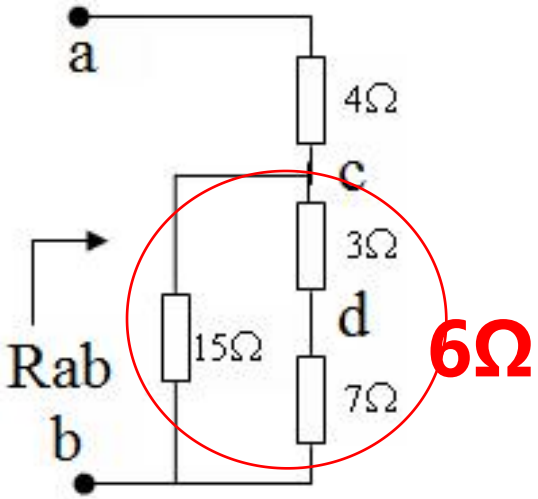
二 二端电阻网络的等效
(equivalent simplification for two-terminal resistor network)

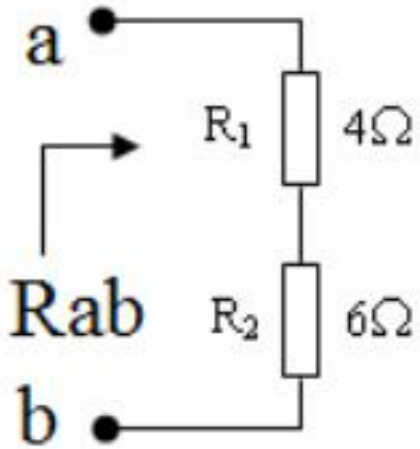
4 电阻串并联例题

例2　求图示电路a、b两端的等效电阻R_{ab}。







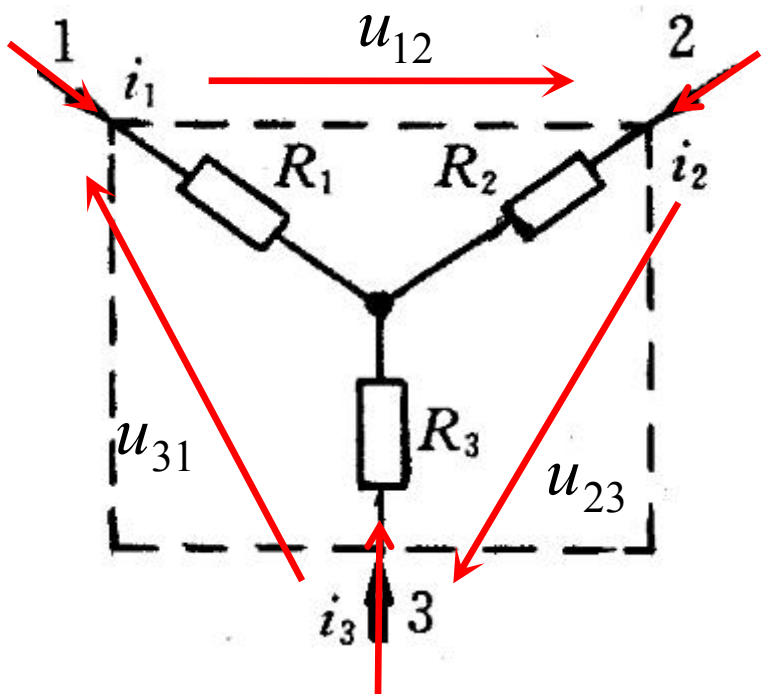


$$R_{ab} = 4 + 6 = 10(\Omega)$$

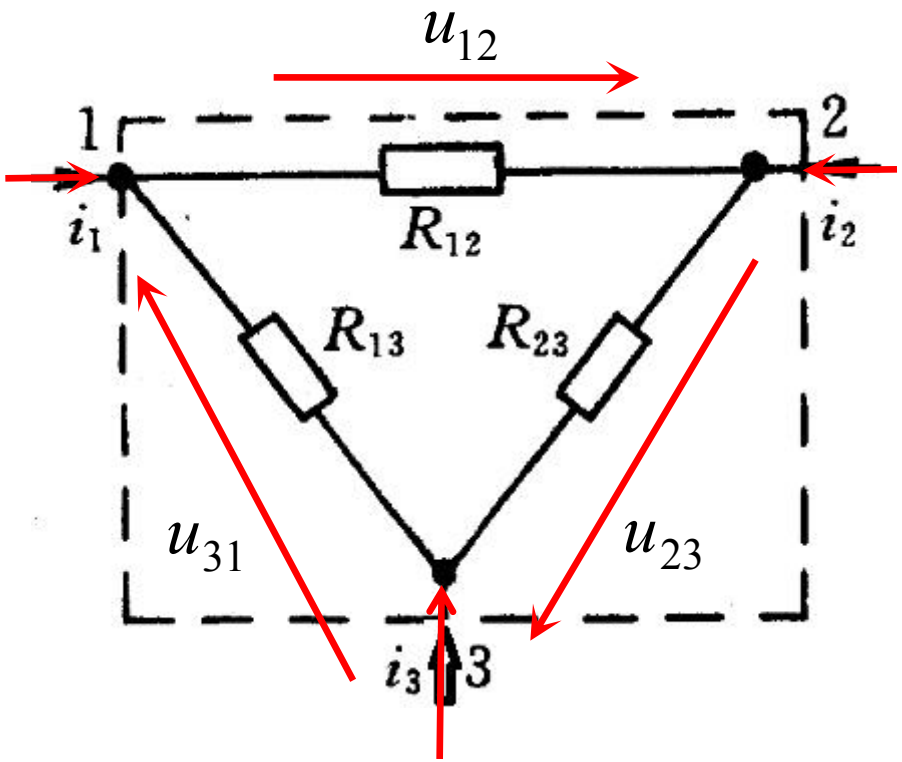
三 → Y形和Δ形电阻网络的等效变换（equivalent transform for Y-Δ resistor network）

1 Y形和Δ形电阻网络的等效变换

Y形网络


1
2
3
u_{12}
i_1
i_2
i_3
R_1
R_2
R_3
u_{31}
u_{23}

Δ形网络

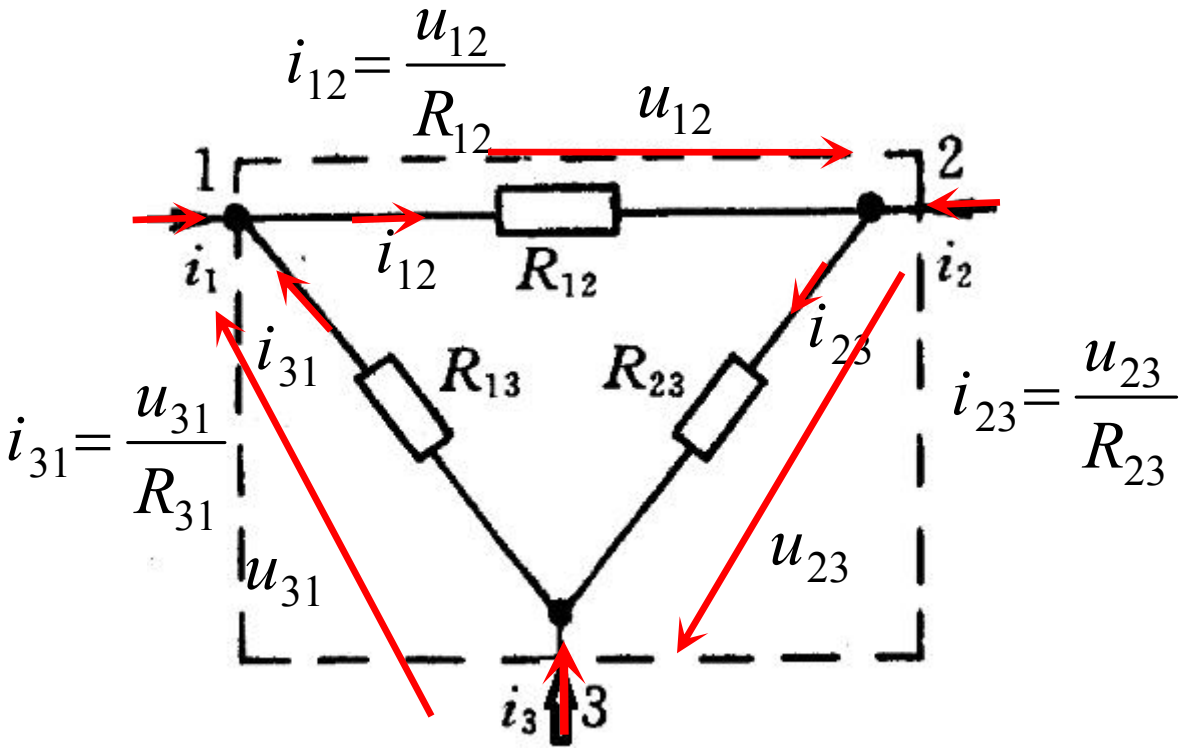

u_{12}
1
2
3
i_1
i_2
i_3
R_{12}
R_{13}
R_{23}
u_{31}
u_{23}

Δ形网络

$$i_1 = i_{12} - i_{31} = \frac{u_{12}}{R_{12}} - \frac{u_{31}}{R_{31}}$$

$$i_2 = i_{23} - i_{12} = \frac{u_{23}}{R_{23}} - \frac{u_{12}}{R_{12}}$$

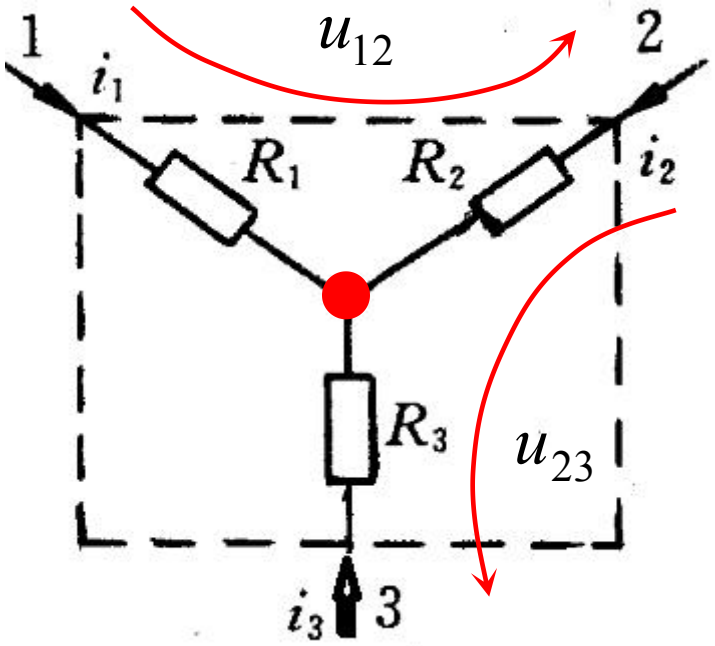
$$i_3 = i_{31} - i_{23} = \frac{u_{31}}{R_{31}} - \frac{u_{23}}{R_{23}}$$



Y形网络

$$\begin{cases} u_{12} = R_1 i_1 - R_2 i_2 \\ u_{23} = R_2 i_2 - R_3 i_3 \\ i_1 + i_2 + i_3 = 0 \end{cases}$$

$$\begin{cases} i_1 = \dfrac{R_3 u_{12}}{R_1R_2 + R_2R_3 + R_3R_1} - \dfrac{R_2 u_{31}}{R_1R_2 + R_2R_3 + R_3R_1} \\ i_2 = \dfrac{R_1 u_{23}}{R_1R_2 + R_2R_3 + R_3R_1} - \dfrac{R_3 u_{12}}{R_1R_2 + R_2R_3 + R_3R_1} \\ i_3 = \dfrac{R_2 u_{31}}{R_1R_2 + R_2R_3 + R_3R_1} - \dfrac{R_1 u_{23}}{R_1R_2 + R_2R_3 + R_3R_1} \end{cases}$$



Y 形网络

$$i_1=\frac{u_{12}}{R_{12}}-\frac{u_{31}}{R_{31}}$$

$$i_2=\frac{u_{23}}{R_{23}}-\frac{u_{12}}{R_{12}}$$

$$i_3=\frac{u_{31}}{R_{31}}-\frac{u_{23}}{R_{23}}$$

$$=$$

$$\begin{cases} i_1=\dfrac{R_3u_{12}}{R_1R_2+R_2R_3+R_3R_1}-\dfrac{R_2u_{31}}{R_1R_2+R_2R_3+R_3R_1} \\ i_2=\dfrac{R_1u_{23}}{R_1R_2+R_2R_3+R_3R_1}-\dfrac{R_3u_{12}}{R_1R_2+R_2R_3+R_3R_1} \\ i_3=\dfrac{R_2u_{31}}{R_1R_2+R_2R_3+R_3R_1}-\dfrac{R_1u_{23}}{R_1R_2+R_2R_3+R_3R_1} \end{cases}$$

Δ 形网络

$$G_{12}=\frac{1}{R_{12}}=\frac{R_3}{R_1R_2+R_2R_3+R_3R_1}=\frac{1/(R_1R_2)}{1/R_3+1/R_1+1/R_2}=\frac{G_1G_2}{G_1+G_2+G_3}$$

$$G_{23}=\frac{G_2G_3}{G_1+G_2+G_3}$$

$$G_{31}=\frac{G_3G_1}{G_1+G_2+G_3}$$

$$\mathrm{Y}\Rightarrow\Delta$$

$$G_\Delta=\frac{\text{Y形相邻电导之积}}{\text{Y形电导之和}}$$

若 $R_1=R_2=R_3=R_Y$

则 $R_{12}=R_{23}=R_{31}=R_\Delta$

$$R_\Delta=3R_Y$$

$$\begin{cases} i_1 = \dfrac{u_{12}}{R_{12}} - \dfrac{u_{31}}{R_{31}} \\ i_2 = \dfrac{u_{23}}{R_{23}} - \dfrac{u_{12}}{R_{12}} \\ i_3 = \dfrac{u_{31}}{R_{31}} - \dfrac{u_{23}}{R_{23}} \end{cases} = \begin{cases} i_1 = \dfrac{R_3 u_{12}}{R_1R_2 + R_2R_3 + R_3R_1} - \dfrac{R_2 u_{31}}{R_1R_2 + R_2R_3 + R_3R_1} \\ i_2 = \dfrac{R_1 u_{23}}{R_1R_2 + R_2R_3 + R_3R_1} - \dfrac{R_3 u_{12}}{R_1R_2 + R_2R_3 + R_3R_1} \\ i_3 = \dfrac{R_2 u_{31}}{R_1R_2 + R_2R_3 + R_3R_1} - \dfrac{R_1 u_{23}}{R_1R_2 + R_2R_3 + R_3R_1} \end{cases}$$

$$\frac{1}{R_{12}} = \frac{R_3}{R_1R_2 + R_2R_3 + R_3R_1}$$

$$\begin{cases} R_{12} = \dfrac{R_1R_2 + R_2R_3 + R_3R_1}{R_3} \\ R_{23} = \dfrac{R_1R_2 + R_2R_3 + R_3R_1}{R_1} \\ R_{31} = \dfrac{R_1R_2 + R_2R_3 + R_3R_1}{R_2} \end{cases}$$

$$\begin{cases} R_1 = \dfrac{R_{12}R_{31}}{R_{12} + R_{23} + R_{31}} \\ R_2 = \dfrac{R_{12}R_{23}}{R_{12} + R_{23} + R_{31}} \\ R_3 = \dfrac{R_{23}R_{31}}{R_{12} + R_{23} + R_{31}} \end{cases}$$

$$\Delta \Rightarrow \mathrm{Y}$$

$$R_{\mathrm{Y}} = \frac{\Delta\text{形相邻电阻之积}}{\Delta\text{形电阻之和}}$$

若 $R_{12} = R_{23} = R_{31} = R_{\Delta}$

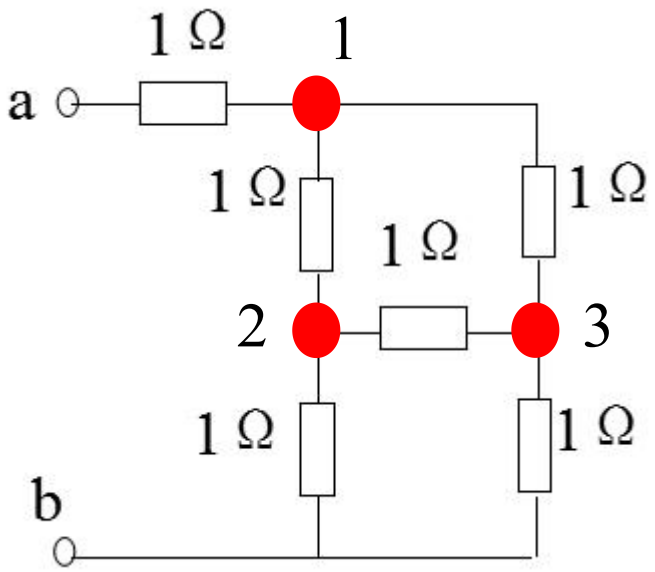
则 $R_1 = R_2 = R_3 = R_{\mathrm{Y}}$

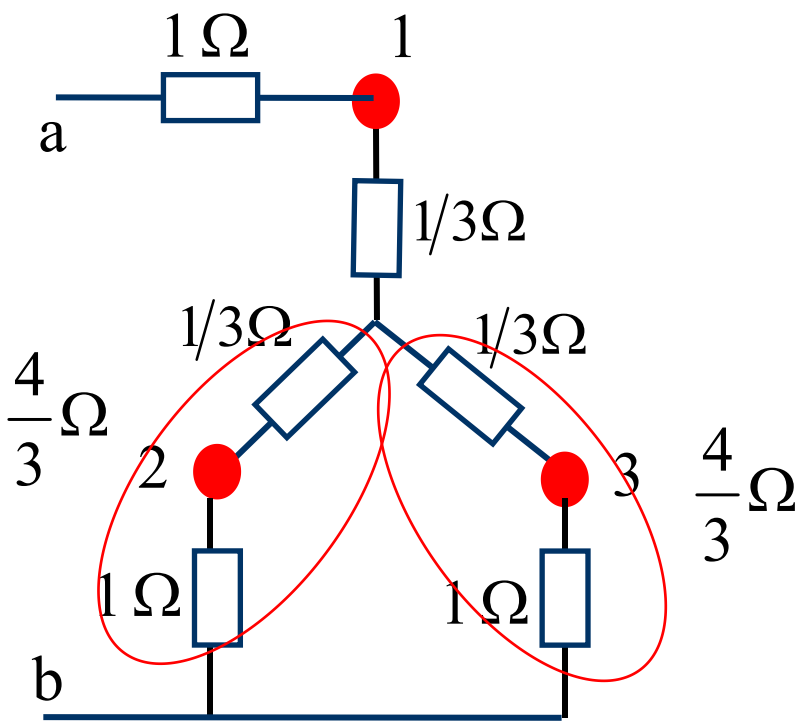
$$R_{\mathrm{Y}} = \frac{1}{3} R_{\Delta}$$

三 → Y形和Δ形电阻网络的等效变换（equivalent transform for Y-Δ resistor network）

2 Y形和Δ形电阻网络等效变换例题

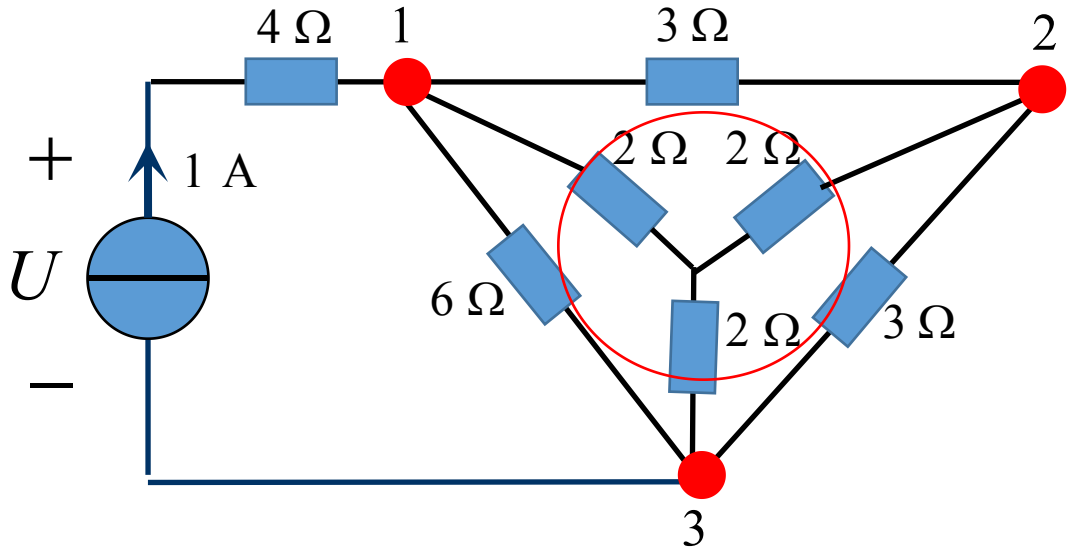
例1 求图示二端网络的输入电阻 R_{ab}

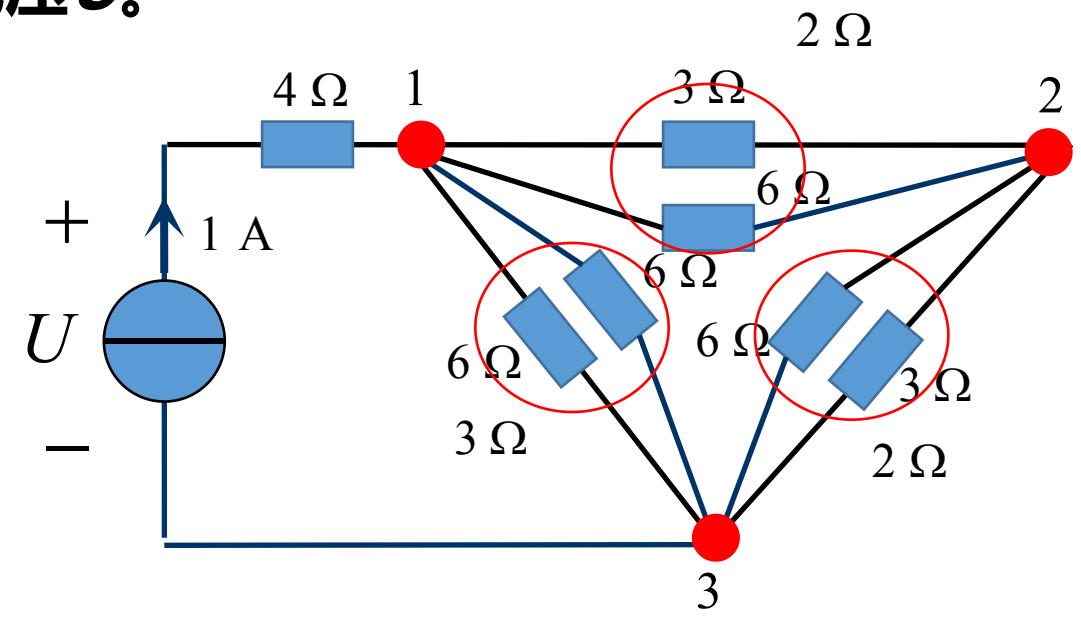




$$R_{ab} = 1 + \frac{1}{3} + (\frac{4}{3} // \frac{4}{3}) = 2\ (\Omega)$$

例2 电路如图所示，求电流源两端的电压*U*。





$$R = 4 + 3//(2+2) = \frac{40}{7}\,(\Omega)$$

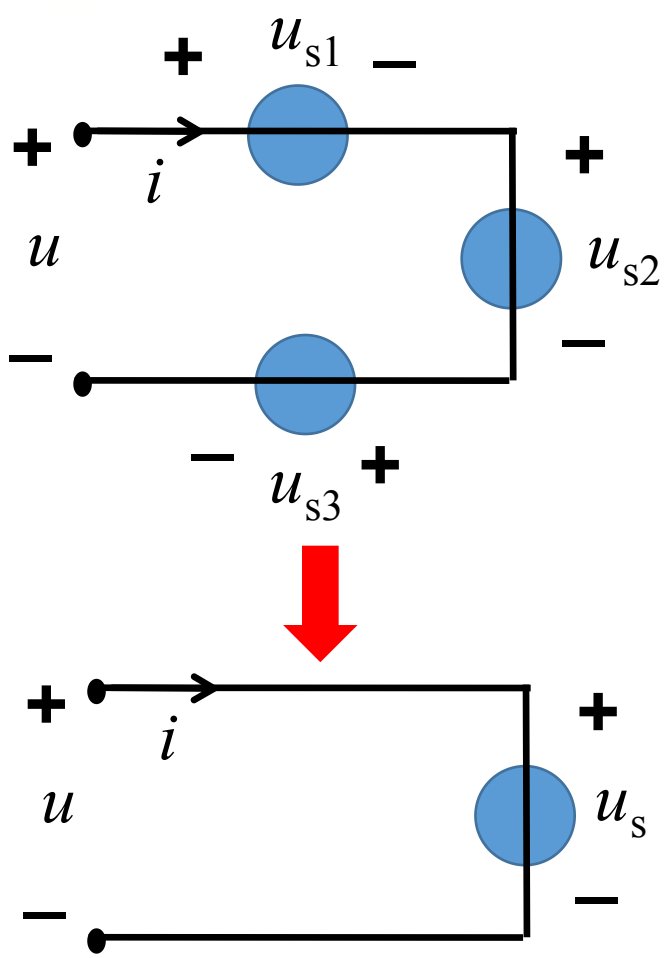
$$U = 1 \times \frac{40}{7} = 5.71\,(\mathrm{V})$$

四 → 二端独立源网络的等效变换（equivalent transform for two-terminal source network）

1 理想电压源网络的等效（equivalent transform for ideal voltage source network）

1. 理想电压源串联

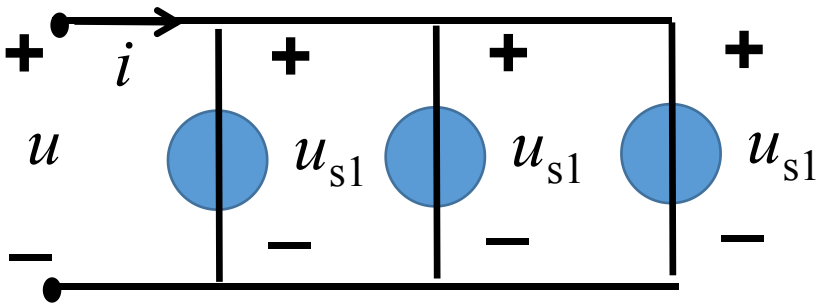
$$u=u_{s1}+u_{s2}+u_{s3}=u_s$$



n个理想电压源串联

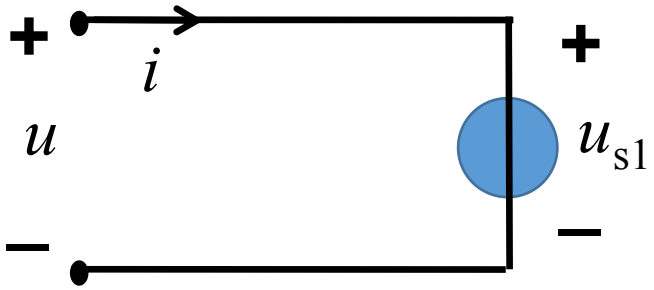
$$u=u_{s1}\pm u_{s2}\pm\cdots\pm u_{sn}$$

2. 理想电压源并联



电压相同的理想电压源并联

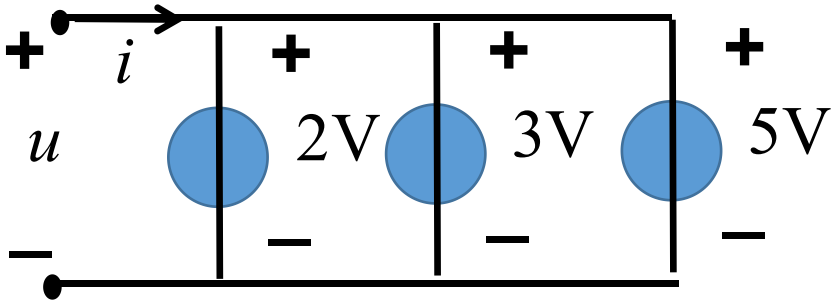
$$u=u_{s1}$$



电压不同的理想电压源，

不可并联 why?

假设真地并联了，
结果如何呢？



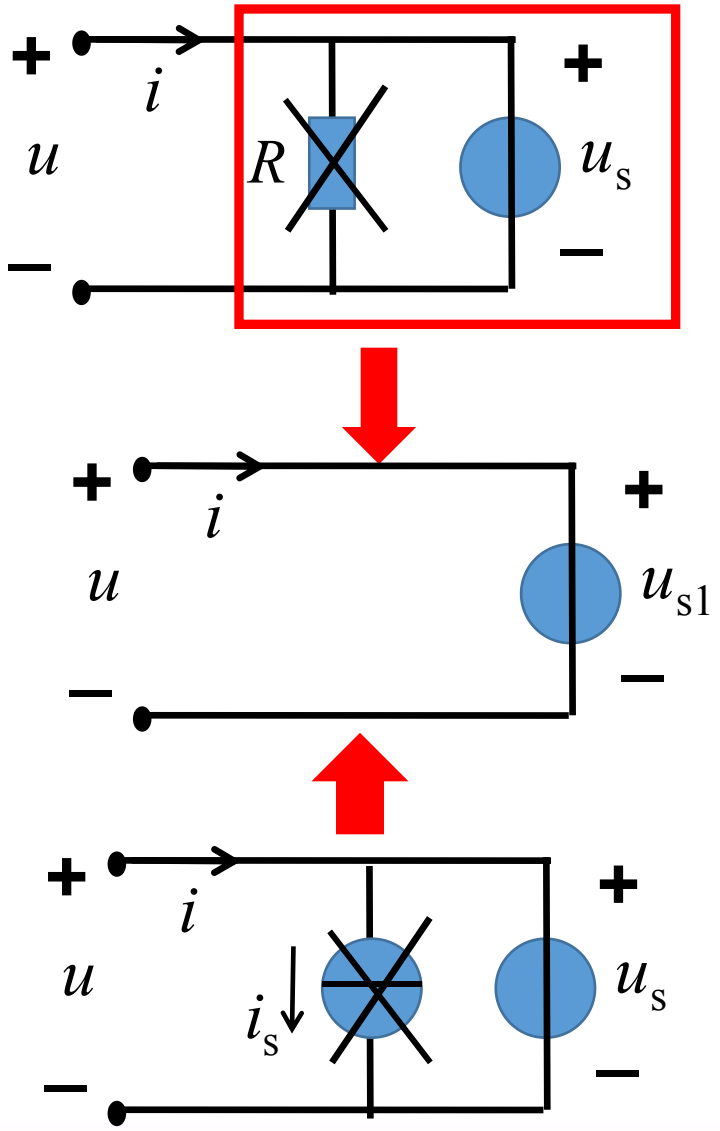
3. 理想电压源与其它并联

理想电压源与电阻并联

$$u=u_s$$

理想电压源与电流源并联

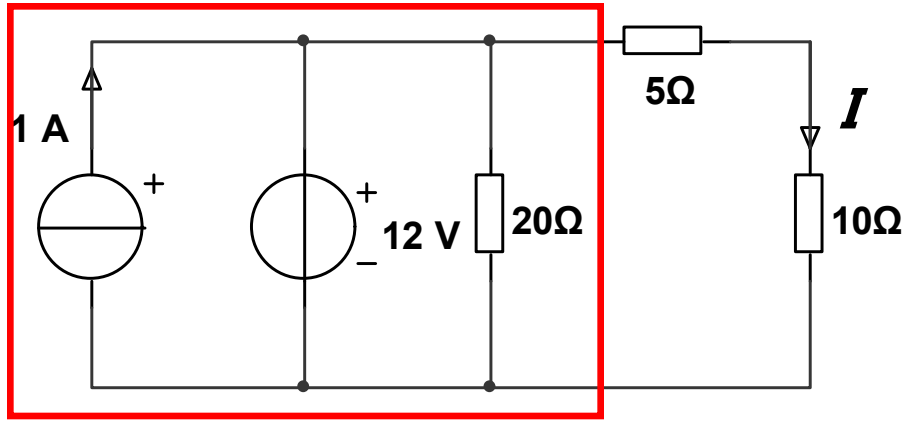
$$u=u_s$$

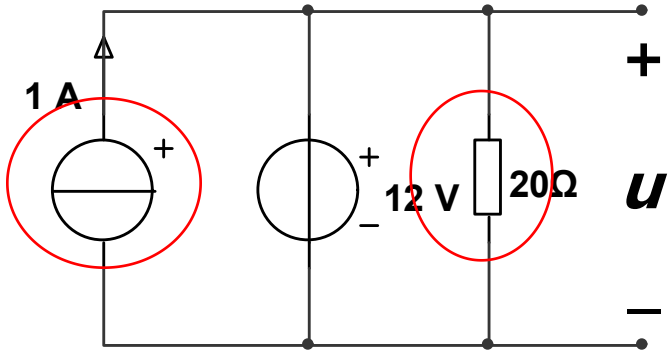


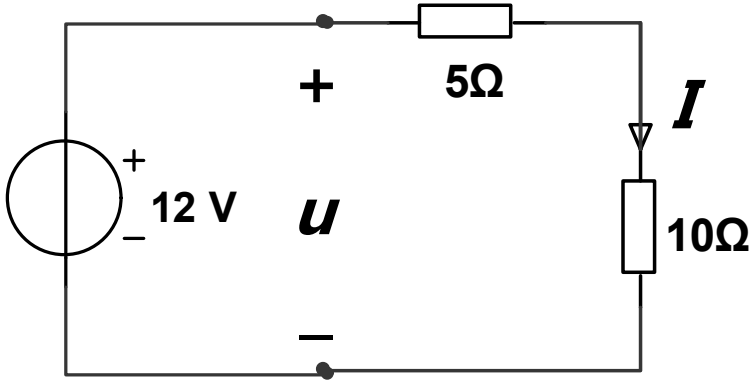
四 → 二端独立源网络的等效变换
（equivalent transform for two-termina source network）

2 理想电压源网络等效例题

例 求图示电路中的电流 I。







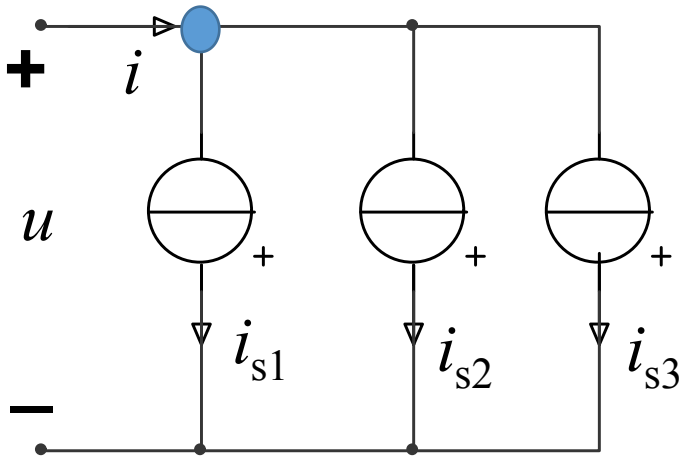
$$I=\frac{12}{5+10}=0.8(\mathrm{A})$$

四 → 二端独立源网络的等效变换（equivalent transform for two-termina source network）

3 理想电流源网络的等效（equivalent transform for ideal current source network）

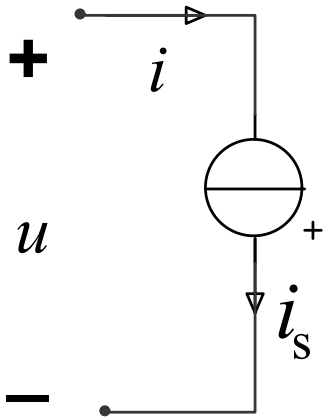
1. 理想电流源并联

$$i = i_{s1} + i_{s2} + i_{s3} = i_s$$

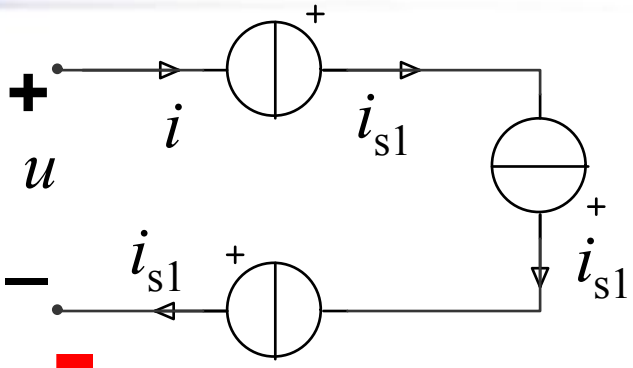


n 个理想电流源并联

$$i = i_{s1} \pm i_{s2} \pm \cdots \pm i_{sn}$$

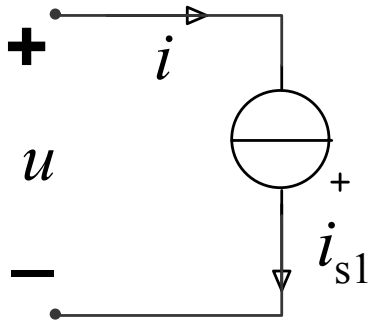


2. 理想电流源串联



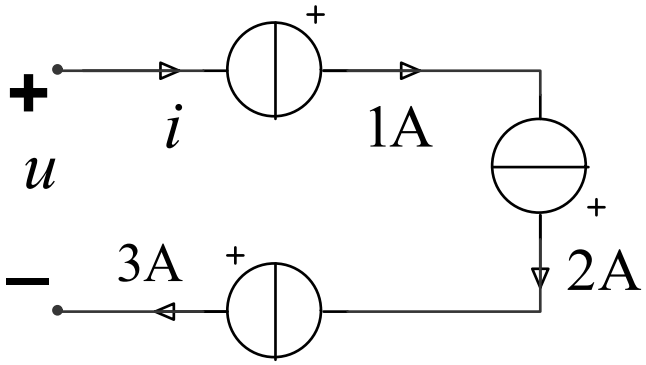
电流相同的理想电流源串联

$$i = i_{s1}$$



电流不同的理想电流源，

不可串联 why?



假设真地串联了，
结果如何呢？

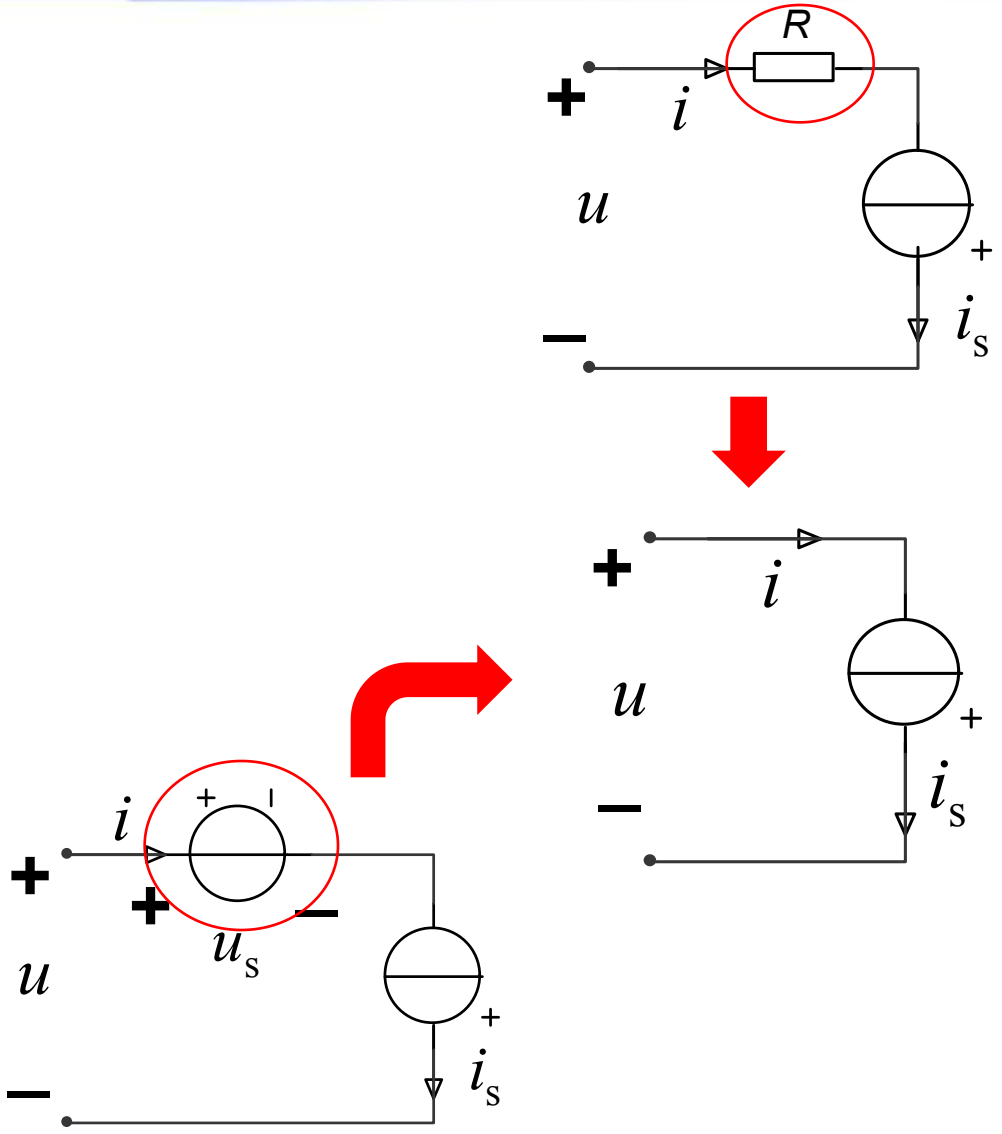
3. 理想电流源与其它串联

理想电流源与电阻串联

$$i = i_s$$

理想电流源与电压源串联

$$i = i_s$$



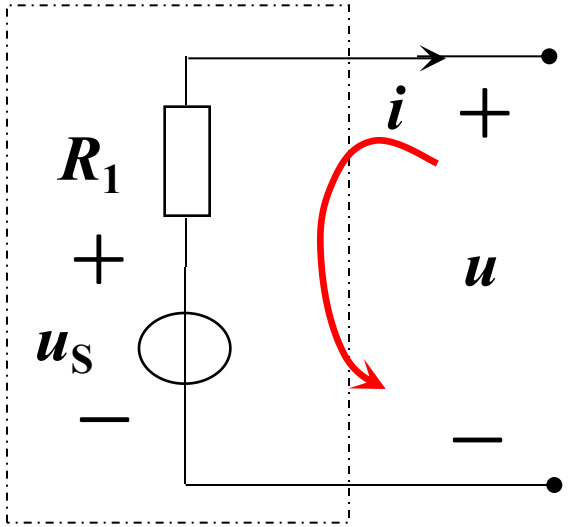
四 → 二端独立源网络的等效变换
（equivalent transform for two-termina source network）

4 有伴电源的相互等效
（equivalent transform for accompanied sources ）

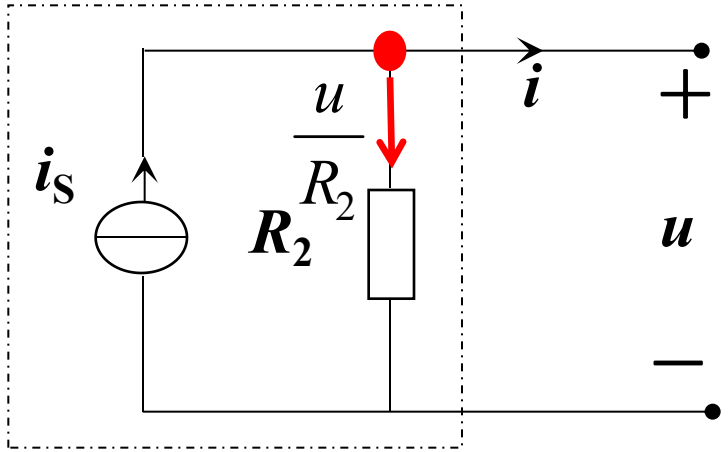
3、 有伴电源的相互等效

有伴电压源

$$u = u_s - R_1 i$$



有伴电流源



$$i = i_s - \frac{u}{R_2}$$

$$u = R_2\, i_s - R_2\, i$$

$$u = u_s - R_1\, i = R_2\, i_s - R_2\, i$$

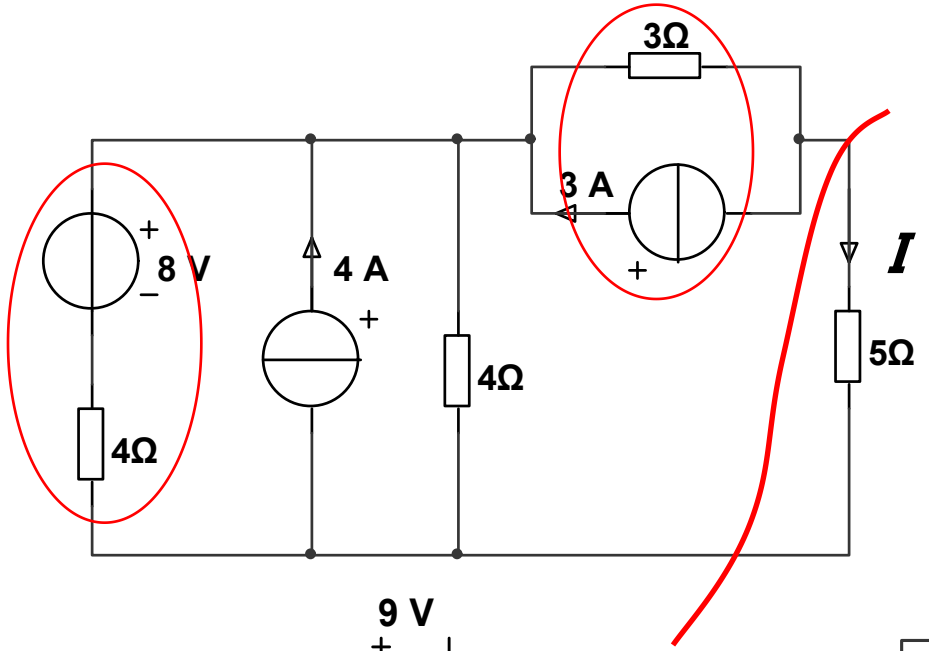
$$u_s = R_2\, i_s \qquad i_s = \frac{u_s}{R_2}$$

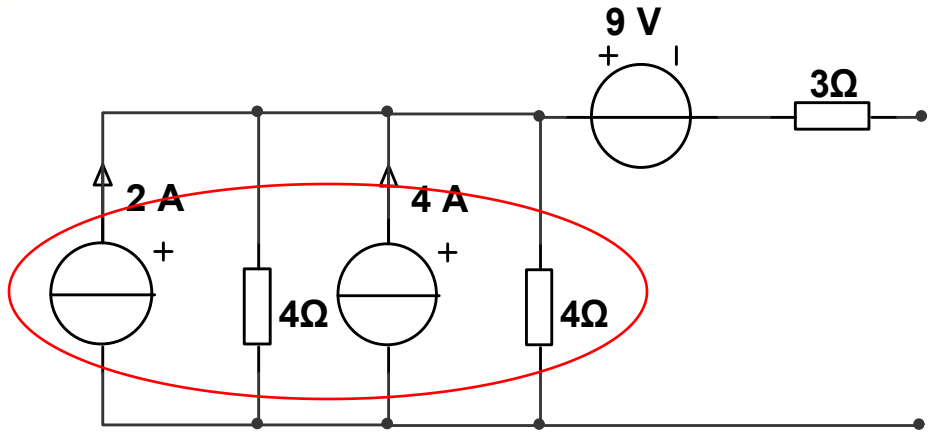
$$R_1 = R_2 = R$$

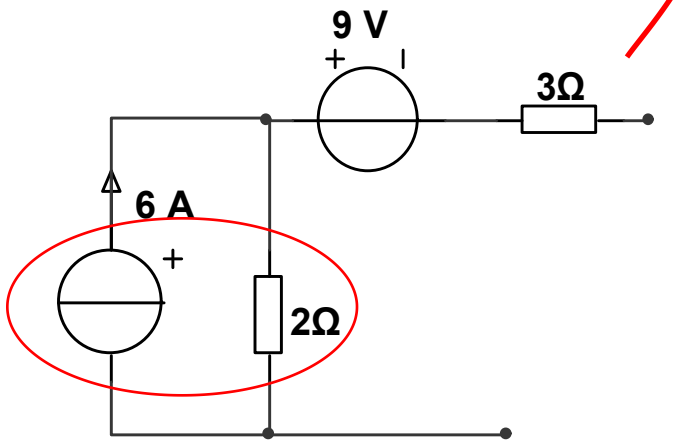
四 → 二端独立源网络的等效变换
（equivalent transform for two-termina source network）

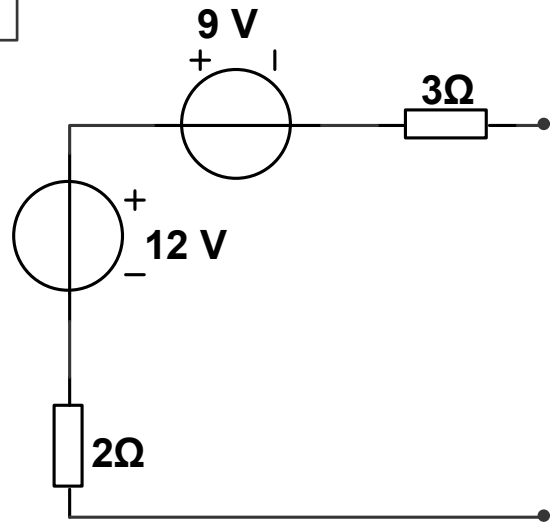
5 有伴电源相互等效例题

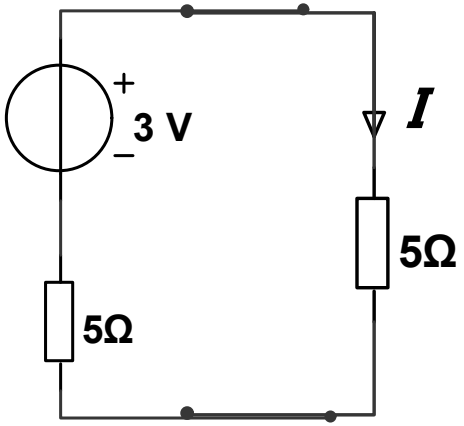
例 求下图电路中的电流 I。











$$I = \frac{3}{5+5} = 0.3\ (\mathrm{A})$$

五 → 含受控源的二端网络的等效
（equivalent transform for two-terminal controlled source network）

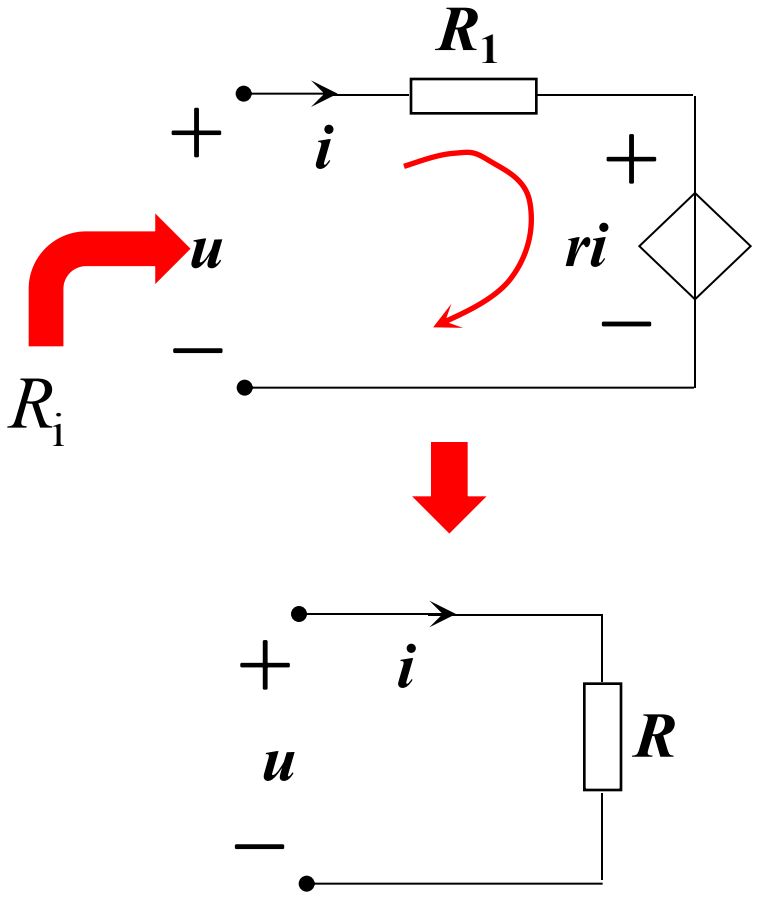
1 含受控源的二端网络的等效

列写端钮的伏安方程

$$u=R_1 i+ri=(R_1+r)i$$

$$R=R_1+r$$

$$u=Ri$$



受控源可等效为电阻

输入电阻

$$R_i=\frac{u}{i}=R_1+r$$

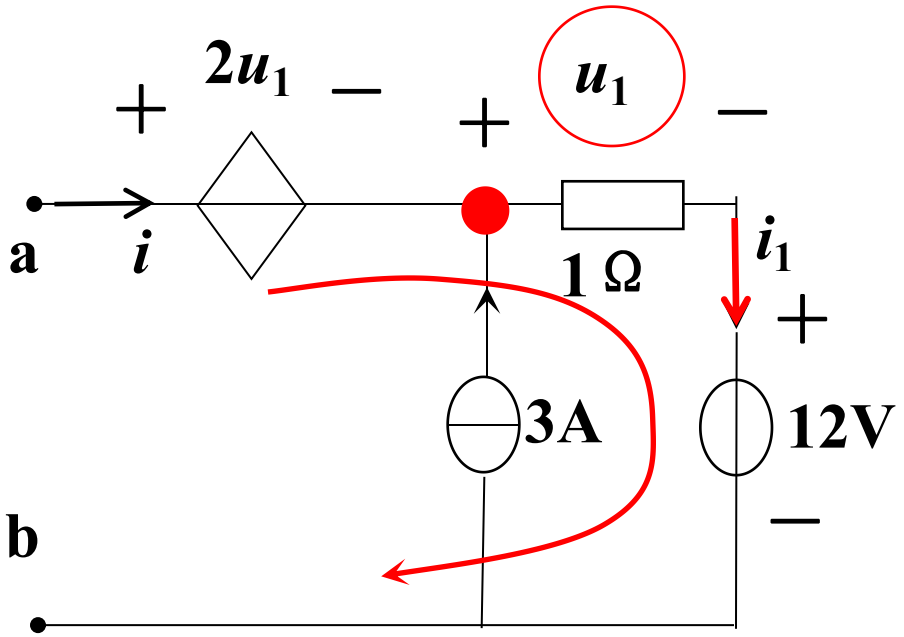
列写端钮的伏安方程

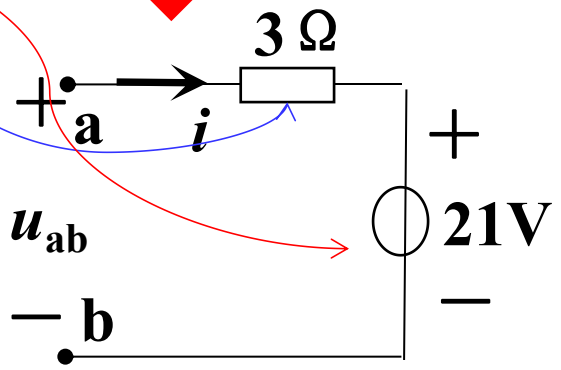
$$u_{ab} = 2u_1 + u_1 + 12$$
$$= 3u_1 + 12$$

$$u_1 = 1 \cdot i_1 = i + 3$$

$$u_{ab} = 3(i+3) + 12 = 3i + 21$$

可等效为有伴电压源





五 → 含受控源的二端网络的等效
（equivalent transform for two-terminal controlled source network）

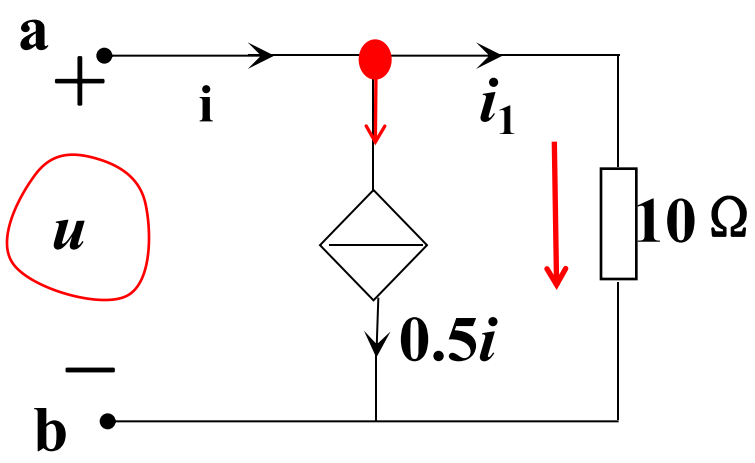
2 含受控源二端网络等效例题

例1 求图示二端网络的输入电阻R_{ab}

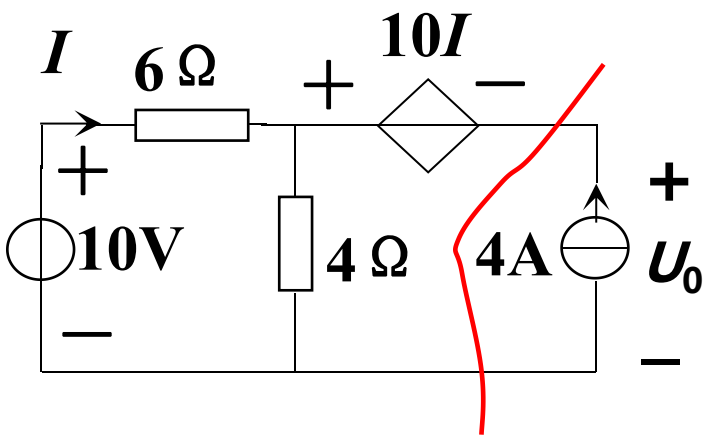
解：

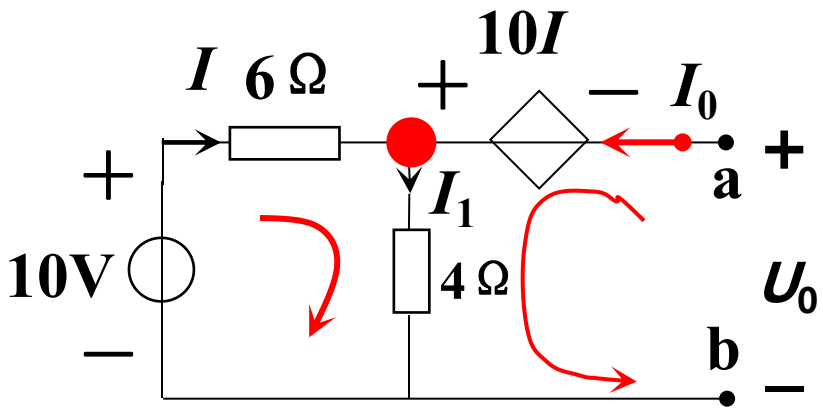
$$u = 10i_1 = 10(i - 0.5i) = 5i$$

$$R_{ab} = \frac{u}{i} = 5(\Omega)$$



例 2 电路如图所示，求4A电流源发出的功率。





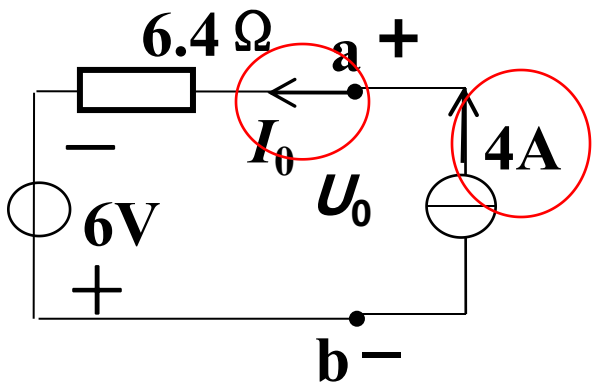
$$U_0 = -10I + 4I_1 = -10I + 4(I + I_0) = -6I + 4I_0$$

$$6I + 4I_1 = 10 \qquad 6I + 4(I + I_0) = 10$$

$$I = 1 - 0.4I_0$$

$$U_0 = -6I + 4I_0 = 6.4I_0 - 6$$

$$= 6.4 \times 4 - 6 = 19.6(\mathrm{V})$$



$$P = -4U_0 = -4 \times 19.6 = -78.4(\mathrm{W})$$

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